SocialJev

A Family of Rapid Classification Foundation Models for the Social Sciences*

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Abstract

We introduce SocialJev, a family of rapid classification foundation models for the social sciences combining the flexibility of modern transformer-based LLMs with the speed of lightweight NLP methods, inspired by TypeSafe's Jev. The models are based on Qwen3-series autoregressive transformers but adapted to return probabilities over arbitrary supplied classes instead of decoding answers autoregressively. The resulting gains in throughput unlock cheap transformer-based text classification at hyper-scale for the social sciences: for instance, one could classify the entirety of English Wikipedia (October 2026) in 6.5 hours on a single GPU, at a rental cost equivalent to two cups of coffee. The model-family is foundational in two senses: 1) their social-science fine-tuning makes it useful without further adaptation for a wide range of tasks, 2) unlike typical classification models, SocialJev returns probabilities over arbitrary classes. A preview-version of the models is available on 🤗 Hugging Face; they can be used with Huggingface's native 'transformers' library or our Python-package, which is available from PyPI and GitHub.

1 Introduction

In order to use text as data for empirical analyses, social scientists apply a wide range of methods to extract structured features from unstructured text (Grimmer and Stewart, 2013; Evans and Aceves, 2016; Gentzkow et al., 2019; Ash and Hansen, 2023). Especially at larger scale, social scientists often face a trade-off: More traditional, lightweight NLP methods are often relatively

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We are still actively developing the models and training methods. If you encounter domains in which they do not yet perform well, please consider reaching out via email or submitting a GitHub issue.

easy to train and inexpensive to run at scale, but they may lack some of the flexibility and subtlety of modern transformer-based Large Language Models (LLMs); frontier LLMs, on the other hand, can often be used after minimal prompt optimisation, but large-scale annotation can be costly (Ash et al., 2026). Relying on closed-source LLMs in particular introduces a separate trade-off, as the eventual retirement of models risks making data pipelines irreproducible in the medium run (cf. e.g., Johnson and Obradovich, 2024; Palmer et al., 2024; Dunn et al., 2026; Wolfrath et al., 2026).

SocialJev Against this backdrop, we introduce SocialJev, a family of rapid classification foundation models for the social sciences, combining the flexibility of modern transformer-based LLMs with the speed of lightweight NLP methods. Inspired by TypeSafe's Jev (Almeida, 2026), the models are built on the autoregressive transformer backbones of the Qwen-3 series, but instead of decoding answer autoregressively (one token at a time), we bypass the standard language model head to return probability distributions over supplied options in one pass, as illustrated in Figure 1. In comparison to typical classification models, SocialJev is "foundational" in two senses:

- Typical classification models are trained to output a probability distribution over fixed, trained classes; SocialJev returns probability distributions over arbitrary supplied classes
- SocialJev is built on transformers trained on internet-scale data and has additionally been fine-tuned on tens of thousands of queries relevant to social-science tasks, making it useful without further adaptation for a wide range of classification problems

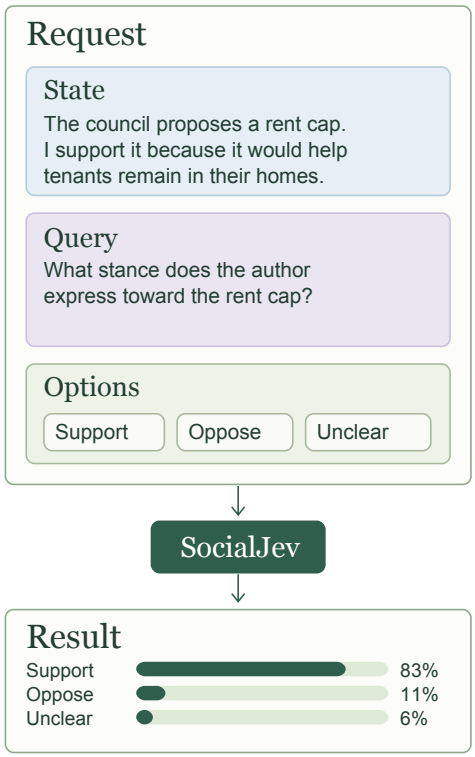


Figure 1: Example classification request and returned probabilities (illustrative).

Decision head. The architecture of TypeSafe's Jev itself is unknown, but over the past few months since its initial release, a wide variety of open-weights/-source implementations have demonstrated comparable quality on emerging public benchmarks (e.g. Kirkby et al., 2026; Chen et al., 2026; Liquid AI, 2026). We evaluate the core variants in our own context but find that both in terms of validation accuracy and their ability to generalise to unseen query types, they are dominated by a simple linear layer and a single-layer MLP at our current scale, if the backbone and decision head are trained jointly.

Data. Our training sample comprises tens of thousands of labelling tasks relevant in social science contexts. We first source the states (contexts) from a wide variety of datasets with a) high-quality human labels, b) a codebook explaining the labels. We then use the human labels as references for a model generating synthetic queries for four core NLP tasks in the social sciences: determining topic/concept-presence, the stance, framing, or sentiment expressed in a given sequence.

Training. We train the backbone and decision head jointly, but optimize them in separate parameter groups with different learning rates, as we find this helps fast convergence and

increases the best validation accuracy. We use simple cross-entropy as objective. We evaluate our models on a held-out split of our data against several alternative models. We find that it performs pareto-optimally and comparable in overall accuracy to frontier LLMs like GLM-5.3 Flash (GLM-5 Team, 2026) or DeepSeek-4.1-Flash (DeepSeek-AI, 2026) while being orders of magnitude smaller, faster, and cheaper to run.

Applications. The SocialJev series of models may unlock large-scale reproducible, simple labelling for social science tasks. We demonstrate this on a random sample of 100k randomly drawn Wikipedia articles using the Python-package we release together with the models. Extrapolating from this sample, SocialJev 0.6B (Preview) would classify the entirety of English Wikipedia in less than 6.5 hours on a single NVIDIA GH200, for the GPU-rental cost of approximately 15 USD – equivalent to about two cups of coffee.

We release three model sizes at 0.6B, 1.7B, and 4B parameters. The weights are openly available on Hugging Face as versioned checkpoints. The models can be used through Hugging Face's native Transformers library or our Python package, available from PyPI and GitHub. Inference is deterministic, subject to floating-point differences. Future releases will expand the training data and evaluation, add extraction with supporting citations, and extend the models to multimodal inputs.

2 Model

The changes introduced by TypeSafe's Jev are easiest understood in contrast to normal transformer-based autoregressive language modelling. For a standard causal transformer, string inputs are first tokenized (turned into numbers the model can understand) and one context independent vector is assigned to each token position. Through the transformer layers, each of the token-representations is then adapted depending on the preceding context. When the final token of the input sequence is processed, the final hidden state of the model is multiplied with the so-called unembedding matrix, containing one vector per token in the model's vocabulary, which returns logits, which are not interpretable in their own right. Applying softmax transforms them into a probability distribution over tokens in the vocabulary, from which one is sampled to begin the response. The sampled token is appended to the prefix, and the same loop continues until certain stopping-tokens are produced.

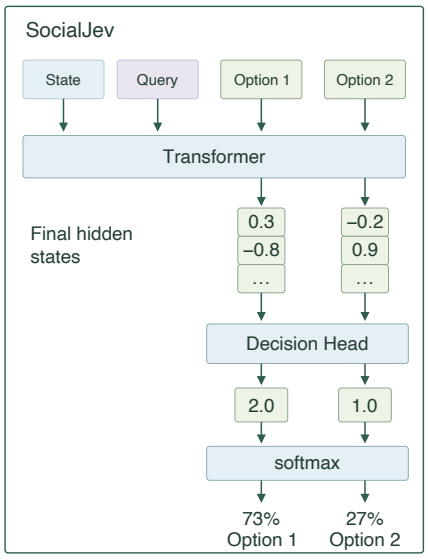


Figure 2: SocialJev architecture. The decision head scores each option's final hidden representation; softmax converts the scores into probabilities. Values are schematic.

Jev-like models work identically, except that their responses are not sampled autoregressively from the backbone. Instead, their so-called *decision-head* directly scores the probability of options supplied at inference time. TypeSafe's implementation is not public, but several open implementations have demonstrated comparable performance on emerging public benchmarks. They differ primarily in how they retrieve scores from the backbone:

- **Contrastive Language Models** encode states and options separately and score their similarity in a learned embedding space (Kwok et al., 2026).

- **d1 (Liquid AI), Mammoth & Elephant (Base Labs)** repurpose the existing language-model head by appending letters to each option, then scoring options by evaluating the probability the model assigns to each letter (Liquid AI, 2026; Kirkby et al., 2026).
- **Clef (Cloudflare)** uses a specialised decision head to score outputs permitted by the supplied schema (Chen et al., 2026).

We implement the core approaches in our own setting and additionally test the following:

- **Linear head:** A single linear layer mapping each option's final hidden representation to a scalar score.
- **MLP head:** A feedforward network with one hidden layer and a nonlinear activation function mapping each option's final hidden representation to a scalar score.

For these two heads, softmax converts the scores into probabilities over the options. We compare the architectures by held-out validation accuracy and their ability to generalise to query types excluded from training. In our setting, when jointly training the backbone and decision head, the linear and MLP variants perform best on both criteria at our current scale. We therefore use linear heads for the released 0.6B and 4B models and an MLP with 512 hidden neurons for the 1.7B model.

Backbone. These heads replace the language-model head of pretrained Qwen3 transformers (Yang et al., 2025), as illustrated in Figure 2. We choose Qwen3 to be able to enforce invariance to query and option order: rearranging the queries associated with a state, or the options associated with a query, does not change their probabilities, since the model does not "perceive" them as alternative options. In Qwen3, we can impose the required separation through attention masks and branch-relative position IDs, as explained in Section 3.4. Newer hybrid backbones such as Qwen3.5 also mix information across tokens through recurrent layers and convolutions (Qwen Team, 2026).

3 Training

3.1 States

We construct training states from 16 datasets covering political topics and positions, media bias, human values, environmental claims, and abusive language. We prioritise datasets with available texts, high-quality human annotations, and well-documented codebooks. The annotations and codebooks provide the basis for generating classification queries relevant to social-science research. Figure 3 shows the distribution of training queries across datasets.

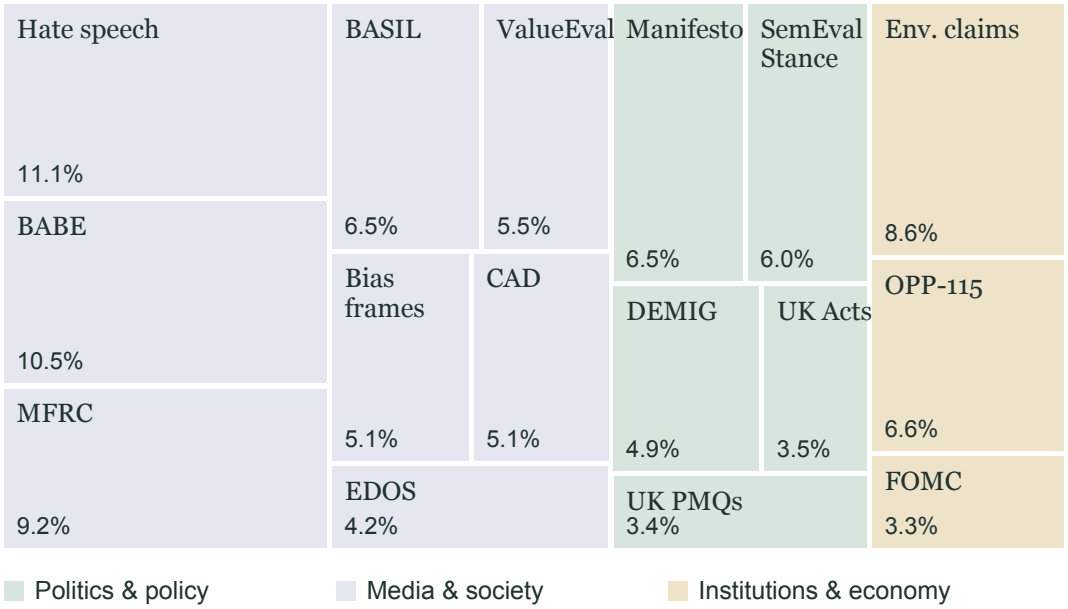


Figure 3: The training mixture. Rectangle areas represent each dataset's share of training questions; colours distinguish broad subject areas.

Note: Sources are OPP-115 (Wilson et al., 2016); Moral Foundations Reddit Corpus (Trager et al., 2022); BABE (Spinde et al., 2021); Social Bias Frames (Sap et al., 2020); CAD (Vidgen et al., 2021); EDOS (Kirk et al., 2023); Environmental Claims (Stammbach et al., 2023); ValueEval (Kiesel et al., 2023); BASIL (Fan et al., 2019); UK Acts and UK PMQs (Bevan and Jennings, 2016; Bevan et al., 2011; Bevan, 2011; John et al., 2013); SemEval Stance (Mohammad et al., 2016); FOMC (Shah et al., 2023); DEMIG (DEMIG, 2015; de Haas et al., 2015); Manifesto Corpus (Lehmann et al., 2026); and Measuring Hate Speech (Sachdeva et al., 2022).

3.2 Queries and Options

Using the states, human annotations, and codebooks as input, we generate training queries and options with Muse-Spark-1.3. The human annotations guide interpretation of the texts, while the codebooks define the concepts originally annotated. We generate queries for four core NLP tasks:

- **Topic presence:** Whether a text discusses a specified topic or concept.
- **Stance:** Whether a text expresses support for or opposition to a specified target.
- **Sentiment:** Whether a text expresses a favourable, unfavourable, or mixed evaluation of a specified target.
- **Framing:** Whether a text presents an issue through a specified interpretive frame.

Researchers differ in how they formulate these tasks, specify their coding instructions, and define the available options. To improve robustness to this variation, the generation prompt encourages diverse wording, answer vocabularies, and formats, ranging from short questions to detailed codebook-style instructions.

3.3 Training objective

We train the pretrained backbone and newly initialised decision head jointly using standard cross-entropy loss over the options. We optimise them in separate parameter groups, allowing their learning rates to be optimised independently.

3.4 Order invariance

We consider order invariance a desirable property of decision models, particularly for scientific applications: rearranging the order of queries or the options within a query should not change the probability assigned to each option. With conventional prompting, these properties are not guaranteed: LLMs exhibit positional biases, including preferences for particular answer identifiers, that can change their predictions when options are reordered (Zheng et al., 2024).

In the spirit of Set-Based Prompting (McIlroy-Young et al., 2024; Brown and McIlroy, 2025), we enforce these properties through the attention mask and positional encoding. Each query attends to the state but cannot attend to other queries. Each option attends to the state and its associated query but cannot attend to other options. Within each branch, attention remains causal. Figure 4 illustrates these restrictions.

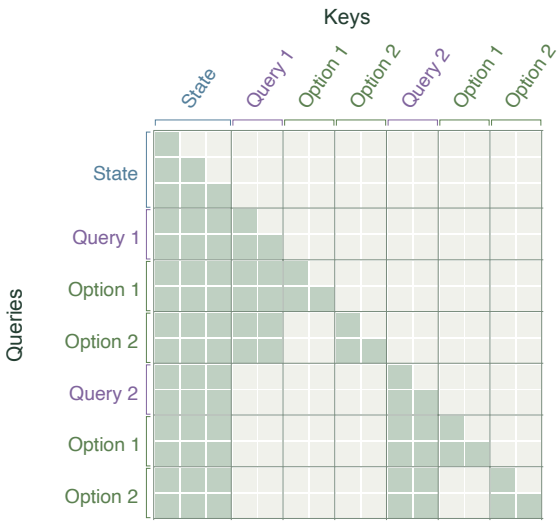


Figure 4: Attention mask enforcing query and option independence. Rows read columns; filled cells permit attention. Axis colours identify states, queries, and options.

The mask prevents information from passing between queries or competing options. We additionally assign position IDs relative to each branch's context, so reordering branches does not change their logical positions. Consequently, each option's score depends on the state, its query, and the option itself, rather than its placement relative to other options. Normalising these scores within each query yields probabilities invariant to query and option order, subject to floating-point differences.

For training efficiency, we also pack multiple examples into fixed-length sequences to reduce padding; the mask additionally blocks attention between examples associated with different states.

3.5 Model selection and evaluation

We select the decision head and checkpoint for each model size by held-out validation accuracy. Table 1 reports the selected heads and their validation scores. Separately, we compare the architectures' ability to generalise to query types excluded from training, as discussed in Section 2.

Table 1: Decision heads selected by validation performance.

Model	Decision head	Val. accuracy
SocialJev 0.6B (preview)	Linear	96.04%
SocialJev 1.7B (preview)	MLP (512 neurons)	97.63%
SocialJev 4B (preview)	Linear	98.58%

We then evaluate the selected checkpoints on the test set across the four task types defined above. Table 2 compares their accuracy with other decision models and general language models (Almeida, 2026; Chen et al., 2026; Liquid AI, 2026; Zaratiana et al., 2025; DeepSeek-AI, 2026; GLM-5 Team, 2026).

Table 2: Test-set classification accuracy overall and by task (percent).

Model	Overall	Topic	Stance	Sentiment	Framing
SocialJev 0.6B (preview)	96.1	98.8	93.5	96.2	94.6
SocialJev 1.7B (preview)	97.5	100.0	94.4	98.1	96.4
SocialJev 4B (preview)	98.0	100.0	96.8	95.3	99.1
TypeSafe Jev 1.13	99.6	100.0	100.0	98.1	100.0
DeepSeek V4.1 Flash	98.8	100.0	99.2	97.2	98.2
GLM 5.3 Flash	98.8	100.0	98.4	98.1	98.2
Clef	96.5	100.0	92.7	92.5	99.1
Clef Flash	93.4	99.4	88.7	90.6	92.0
Liquid d1-3B	90.0	98.8	85.5	82.1	89.3
GLiNER2.5-Decide	62.3	67.6	60.5	45.3	72.3
GLiNER2.5-multi-Decide	51.6	61.8	32.3	40.6	67.9

Overall test accuracy increases from 96.1 percent for SocialJev 0.6B to 98.0 percent for SocialJev 4B. The latter trails DeepSeek and GLM by 0.8 percentage points and Jev by 1.6 points, while being substantially smaller than DeepSeek and GLM and likely smaller than Jev, whose parameter count is estimated rather than disclosed (JevResearch, 2026). Figure 5 illustrates this accuracy–size tradeoff. The preview models thus approach frontier-model accuracy while remaining small enough to host inexpensively on university or rented hardware; unlike Jev, they can be deployed independently of a hosted service.

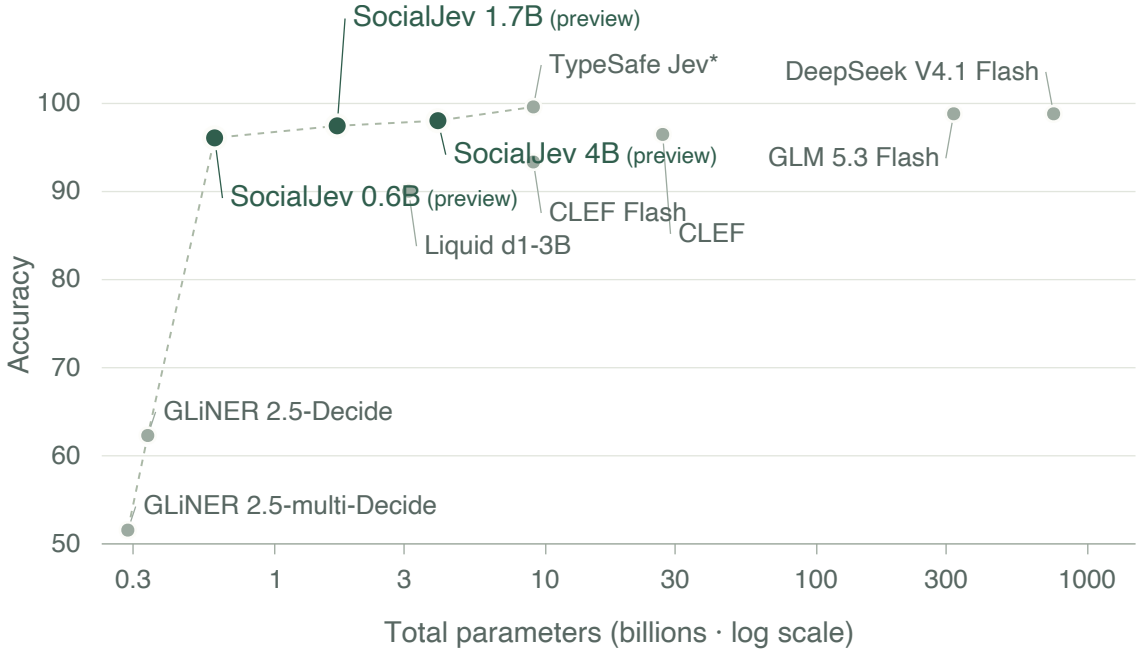


Figure 5: Classification accuracy and model size. Green points identify SocialJev; the dashed line marks the Pareto frontier. Jev is plotted at 9B parameters, the upper end of JevResearch's estimated 4–9B range under a dense-model interpretation (JevResearch, 2026).

4 Applications

The speed and compact size of SocialJev make transformer-based annotation accessible for large social-science corpora. We demonstrate this by labelling 100,000 English Wikipedia articles randomly sampled from the October 2026 snapshot of Wikipedia Monthly (Kamali and Omneity Labs, 2025), using the inference package we supply for large-scale workloads with a vLLM backend.

To assess the throughput and cost of self-hosted annotation, we classify 100,000 English Wikipedia articles randomly sampled from the October 2026 snapshot of Wikipedia Monthly (Kamali and Omneity Labs, 2025). We split long articles into passages of up to 4,096 tokens and ask one binary question per passage: "Does this passage discuss education policy?" All three models process the same material on one NVIDIA GH200, using the vLLM backend with 16 concurrent asynchronous requests, each containing a batch of 128 passages. The maximum input length is 8,192 tokens. Preparation and model startup are excluded from the timed run.

Figure 6 shows document and token throughput. SocialJev 0.6B processes approximately 310 articles per second, compared with 160 for SocialJev 1.7B and 68 for SocialJev 4B. Extrapolating the 0.6B run to all English Wikipedia gives approximately 6.5 hours on a single GH200, at an estimated rental cost of 15 US dollars, equivalent to the price of about two cups of coffee.

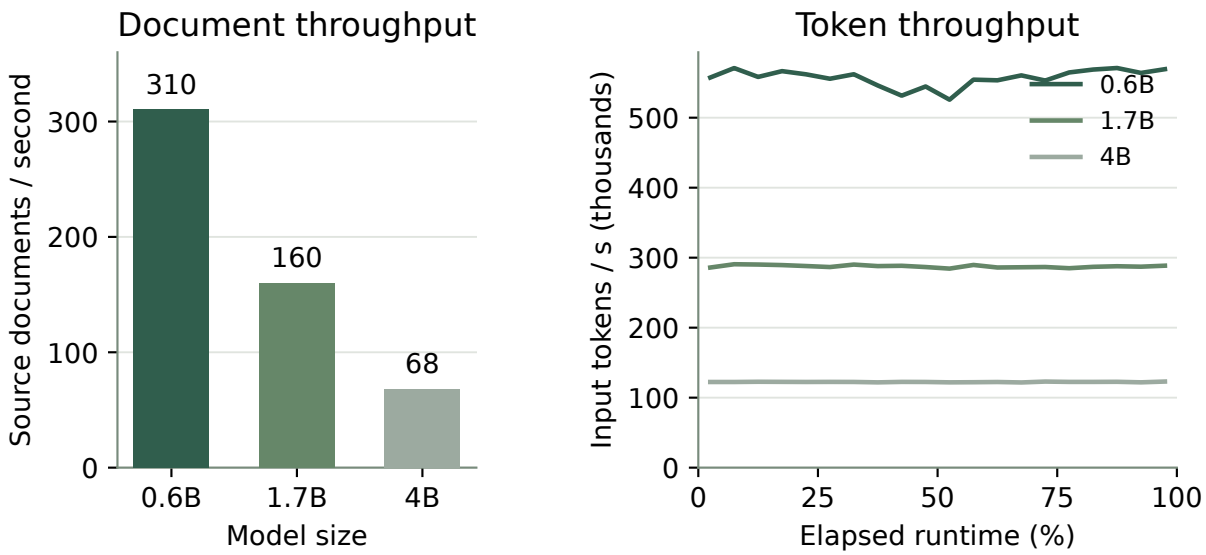


Figure 6: Wikipedia inference throughput on one GH200. Left: articles per second. Right: input tokens per second, including cached prefixes, over normalised elapsed runtime; lines show rates in five-percent intervals. Preparation and startup are excluded.

5 Roadmap

This report documents the preview models released on Hugging Face. We are improving all aspects of the pipeline and package, in particular:

1. Expanding the training data and scaling the models.
2. Adding extraction with citations to supporting text and simple numerical features
3. Extending the models to multimodal inputs.
4. Integrating GPU providers to simplify deployment.
5. Providing interfaces for fine-tuning, evaluation, and query optimisation.

6 Conclusion

We introduce SocialJev to combine the flexibility of LLMs with the speed and low cost of traditional NLP methods, retaining reproducibility through open, versioned weights. By bypassing autoregressive decoding, SocialJev returns probabilities over arbitrary options directly, rather than requiring fixed output classes and dimensions. Fine-tuning across social-science tasks is intended to make the models useful without further task-specific adaptation.

The preview models approach the accuracy of substantially larger models while remaining small enough to host inexpensively on university or rented hardware. Together, these properties make large-scale transformer-based annotation accessible without dependence on a closed-weight provider's continued service. We currently release three model sizes and an inference package and are extending the functionality of both the package and training pipeline.

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